Data Science & Artificial Intelligence

Machine Learning

Support Vector Machines

DPP 01 Discussion Notes

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[MCQ]

#Q. Consider the dataset with six datapoints: $\{(x_1, y_1), (x_2, y_2), \ldots, (x_6, y_6)\}$, where $x_1 = \begin{bmatrix}1\\0\end{bmatrix}$, $x_2 = \begin{bmatrix}0\\1\end{bmatrix}$, $x_3 = \begin{bmatrix}0\\-1\end{bmatrix}$, $x_4 = \begin{bmatrix}-1\\0\end{bmatrix}$, $x_5 = \begin{bmatrix}2\\2\end{bmatrix}$, $x_6 = \begin{bmatrix}-2\\-2\end{bmatrix}$ and the labels are given by $y_1 = y_2 = y_5 = 1$, and $y_3 = y_4 = y_6 = -1$. A hard margin linear support vector machine is trained on the above dataset.

Which ONE of the following sets is a possible set of support vectors?

PYQ

• done in class

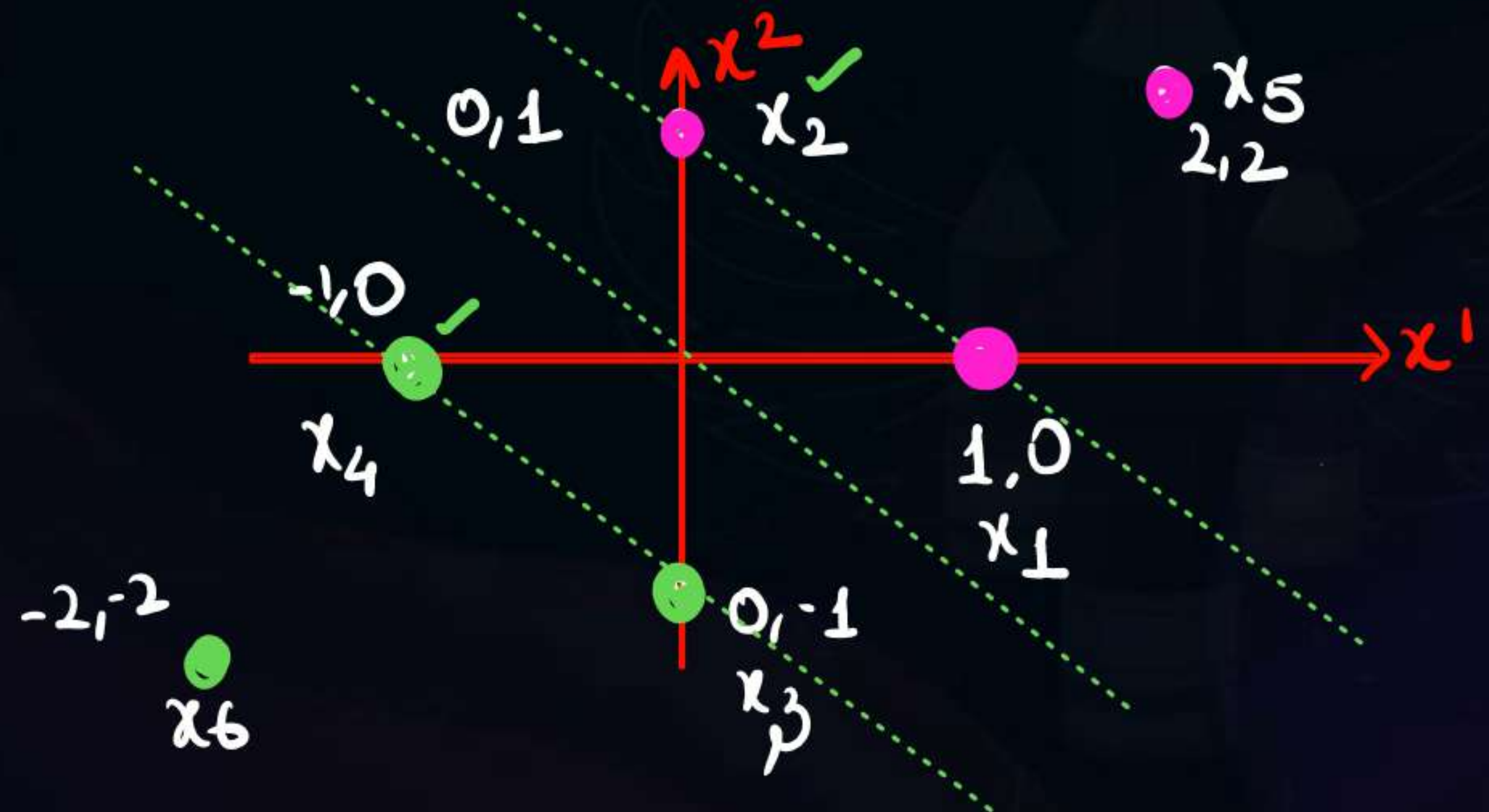
A $\{x_1, x_2, x_5\}$

4 SV

B $\{x_3, x_4, x_5\}$

C $\{x_4, x_5\}$

D ✓ $\{x_1, x_2, x_3, x_4\}$



[MCQ]

#Q. In an SVM, what are support vectors?

A) Data points that lie closest to the decision boundary

B) Data points from all classes

C) Data points that are outliers

D) Data points with the highest dimensionality

[MCQ]

#Q. In a linearly separable dataset, the optimal hyperplane in SVM is the one that:

A Passes through the center of the data points

B $\boxed{\text{Maximizes the distance between support vectors}}$ → max distance b/w SV's
max margin

C Minimizes the distance between support vectors

D Creates the largest number of support vectors

[MCQ]

#Q. What is the advantage of SVM over other classification algorithms?

A. It is immune to overfitting → RBF kernel overfit ✓

B. It works only on linearly separable data

C. It always performs well with unbalanced datasets → work well,

D. It is effective in high-dimensional spaces and nonlinear data ✓

[MCQ]

• Point on margin is SV

#Q. Pick the correct option from the below two statements:

S1: Consider a point which is correctly classified and is on the margin if we remove the point and the decision boundary changes then it is a support vector.

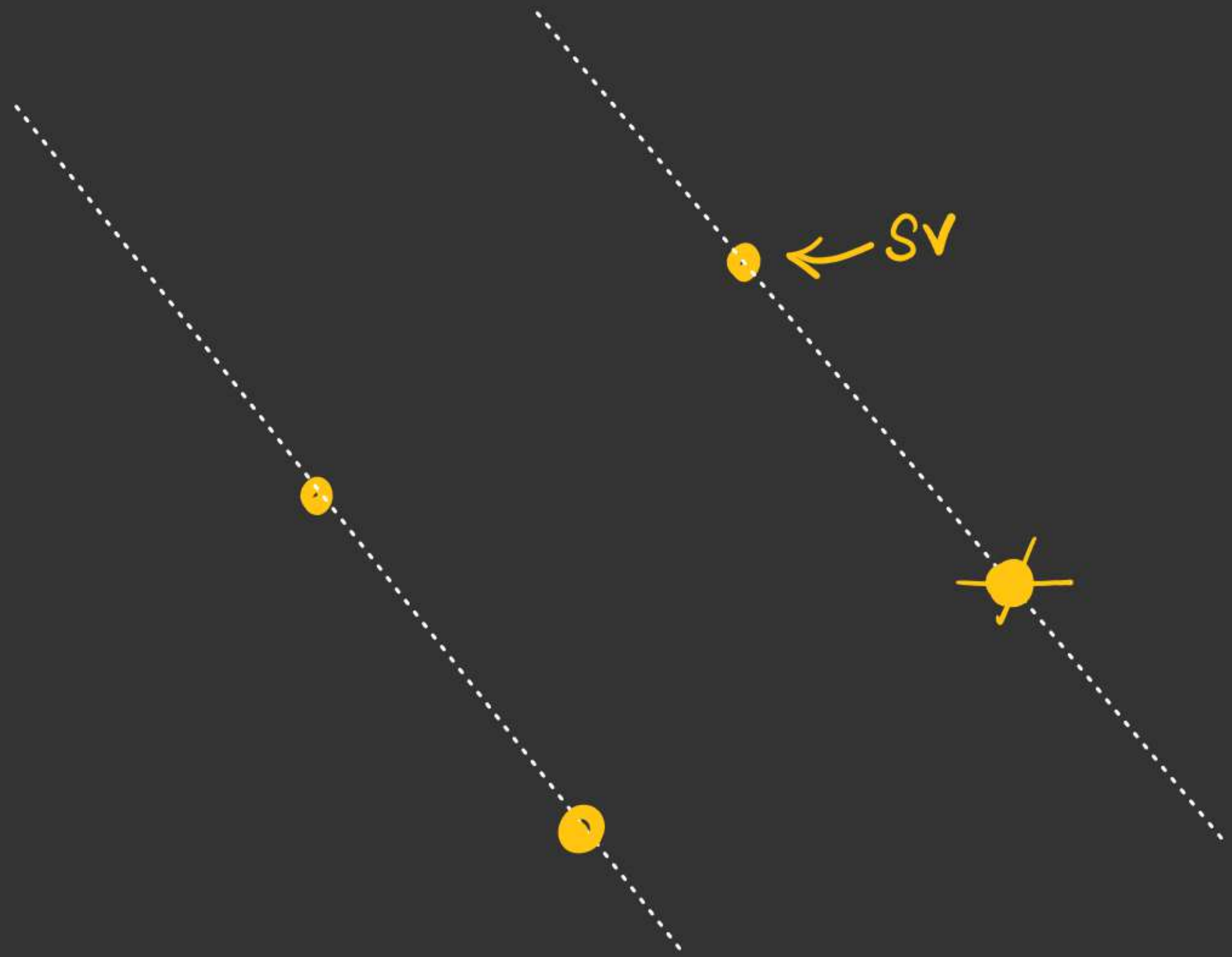
S2: Consider a point which is correctly classified and is on the margin if we remove the point and decision boundary doesn't change then it is not a support vector.

A Only S1 is true

B Only S2 is true

C Both are true

D Both are false


SV

[NAT]

#Q. Find the decision boundary given by Hard-Margin SVM.

-8

1 Class 0

Class 1

decision boundary

$x_1 + 3.5 = 0$

$x_1 = -8$

$x_1 = 1$

$x_1 = -3.5$

[NAT]

#Q. Every trained two-class hard-margin SVM model has at least one point of each class at a distance of exactly $\frac{1}{||w||}$ (the margin width) from the decision boundary. (Please enter 1 for True and 0 for false)

Yes

True ✓

[MCQ]

#Q. Suppose we have a soft-margin SVM for which $\boxed{w = (-6, 3) \text{ and } b = 5.}$ Consider the example $\boxed{(x = (2, 1), y = -1).}$

(I) How does the SVM classify x?

(II) Does (x, y) satisfy the SVM constraint?

Mark the correct option for statement (I) and (II) respectively.

A. −1, Yes

B. +1, Yes

C. −1, No

D. +1, No

$w_1x_1 + w_2x_2 + b$

$-6\times 2 + 3\times 1 + 5$

$\Rightarrow -12 + 3 + 5$

$\Rightarrow -4$

• Point is below classifier ⇒ Class -1

[MCQ]

#Q. Which of the following statements is true regarding the slack variable ξ in SVM classification?

• $\xi \geq 0$

$0 < \xi < 1 \rightarrow$ lie inside margin, classified correctly

A ✗ $\xi > 0$ indicates that the point is classified correctly but lies inside the margin.

B ✗ $\xi = 0$ indicates that the point lies on the decision boundary. ✗

C ✓ $\xi > 1$ indicates that the point is misclassified. ✓

D ✗ $\xi < 0$ indicates that the point is classified correctly but outside the margin.

[MCQ]

#Q. At the optimal solution of

$$\min_{b,w,\xi} \frac{1}{2} w^T w + C.\sum_{n=1}^{N} \xi_n$$

Simple

s. t. $y_n(w^T z_n + b) \geq 1 - \xi_n$ and $\xi_n \geq 0$ for all n,

assume that $y_1(w^T z_1 + b) = -10$. What is the corresponding ξ_1?

A 1

B 11 ✓

C 9

D 31

[MCQ]

#Q. Which is the possible value of ξ_n for point B?

A $\xi_n = 1$

B $\xi_n < 1$

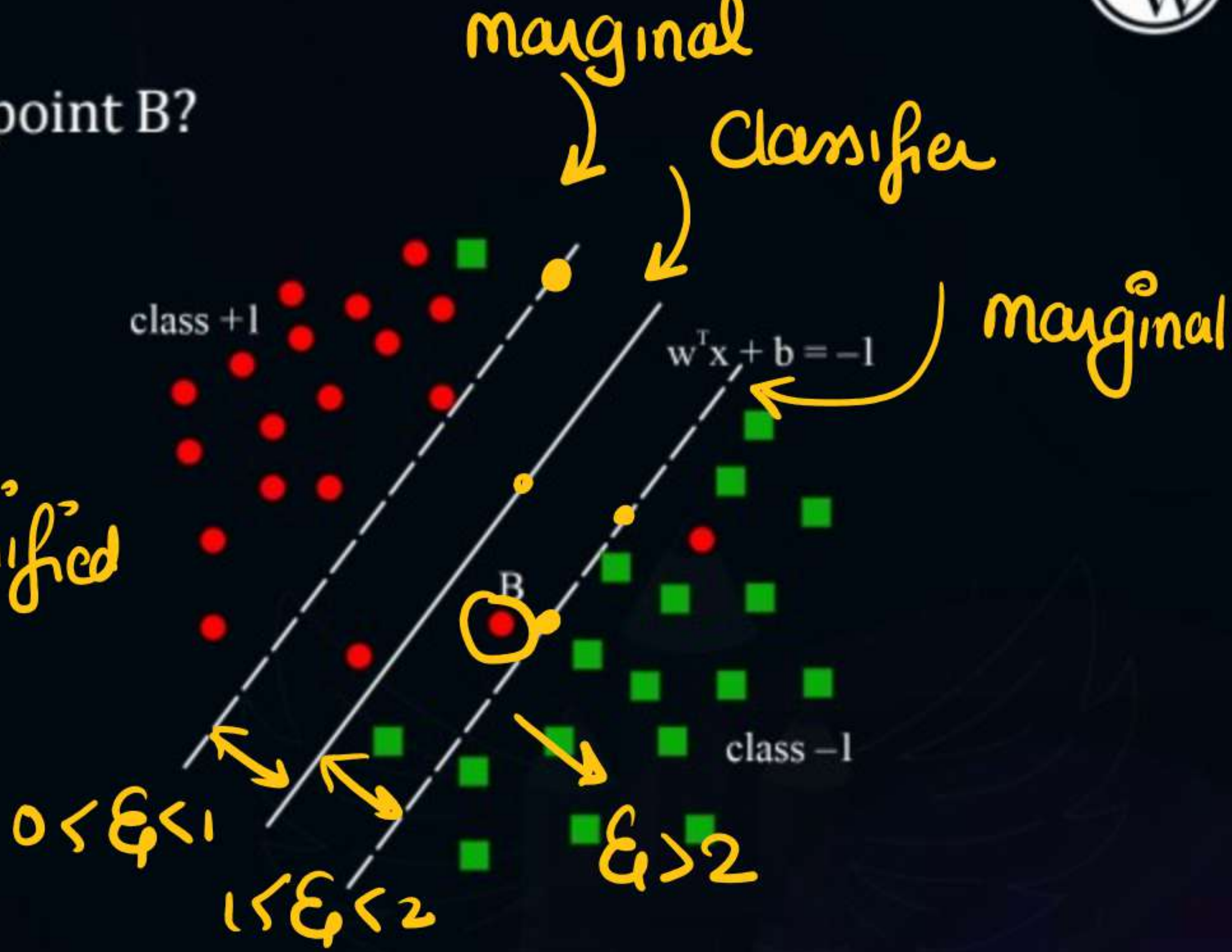
C $1 < \xi_n < 2$

D $\xi_n > 2$

• B point Class +1

• B is wrongly Classified

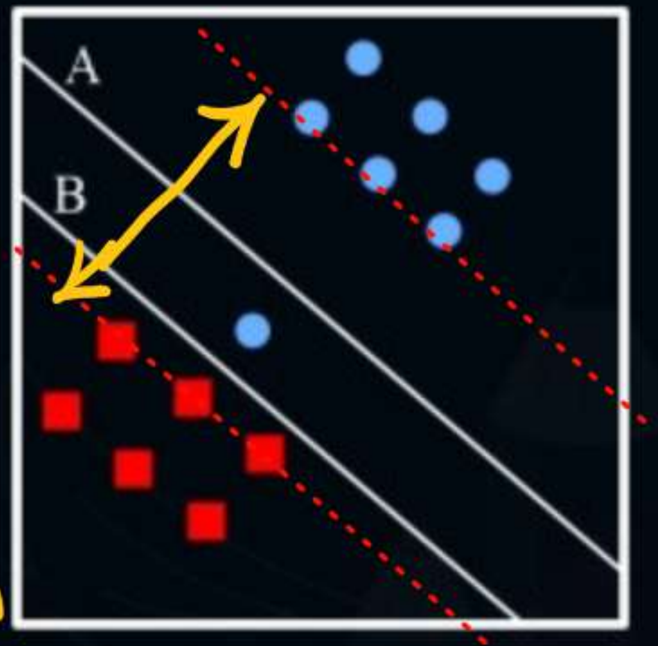
• $\underline{\underline{\xi_i > 1}}$



[NAT]

#Q. Which decision boundary do you prefer if little misclassifications are allowed?

(Please enter 1 for A and –1 for B)



- Clanifier B → Zero error
  margin poor

- we Choose A ✓ → some misclanification allowed, margin will be large.

[MCQ]

#Q. What are support vectors:

A The examples farthest from decision boundary.

B $\boxed{\text{The only examples necessary to compute f(x) in an SVM.}}$ ⇐ Perfect

C The class centroids.

D All the examples that have a non-zero weight α_k in a SVM.

[MCQ]

#Q. Suppose an SVM has $w = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $b = -1$. What is the actual distance between the two planes defined by $w^T x + b = -1$ and $w^T x + b = 1$?

A $\frac{1}{\sqrt{14}}$

B $\frac{2}{\sqrt{14}}$

C $\frac{\sqrt{11}}{2}$

D $\sqrt{11}$

$\frac{2}{\sqrt{1^2+2^2+3^2}}$

$\frac{2}{\sqrt{14}}$

[MCQ]

#Q. If a hard-margin support vector machine tries to minimize $|w|^2$ subject to $y_i(X_i.w + \alpha) \geq 2$ instead, what will be the width of the slab (the point-free region bracketing the decision boundary)?

$$\left.\begin{aligned} wx_i+b &= 2 \\ wx_i+b &= -2 \end{aligned}\right\}$$

A $\frac{1}{||w||}$

B $\frac{4}{||w||}$

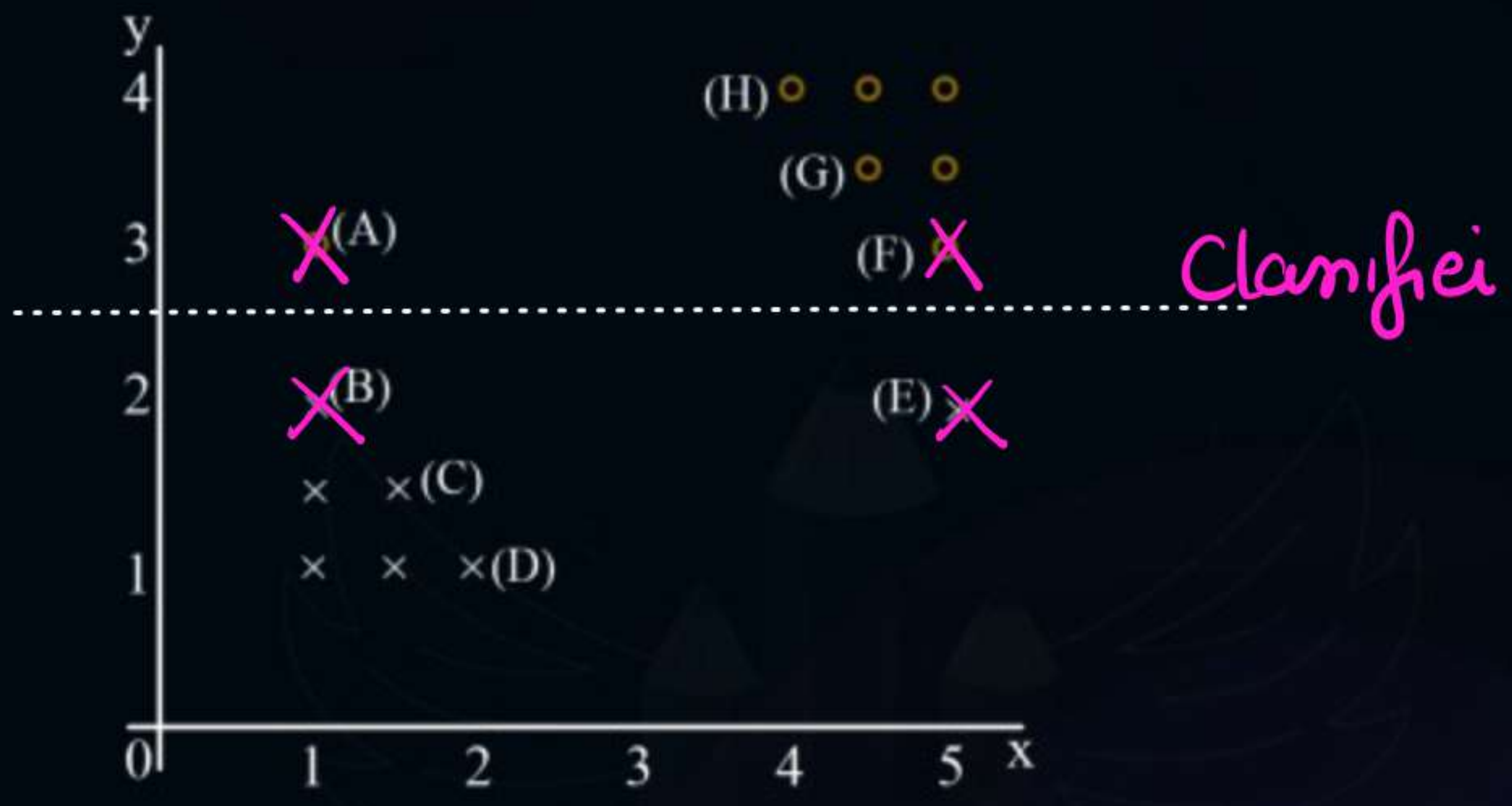
C $\frac{2}{||w||}$

D $\frac{1}{2||w||}$

distance $\frac{4}{||w||}$

[MCQ]

#Q. Select all points that are support vectors. (Hard margin)



A ABEF

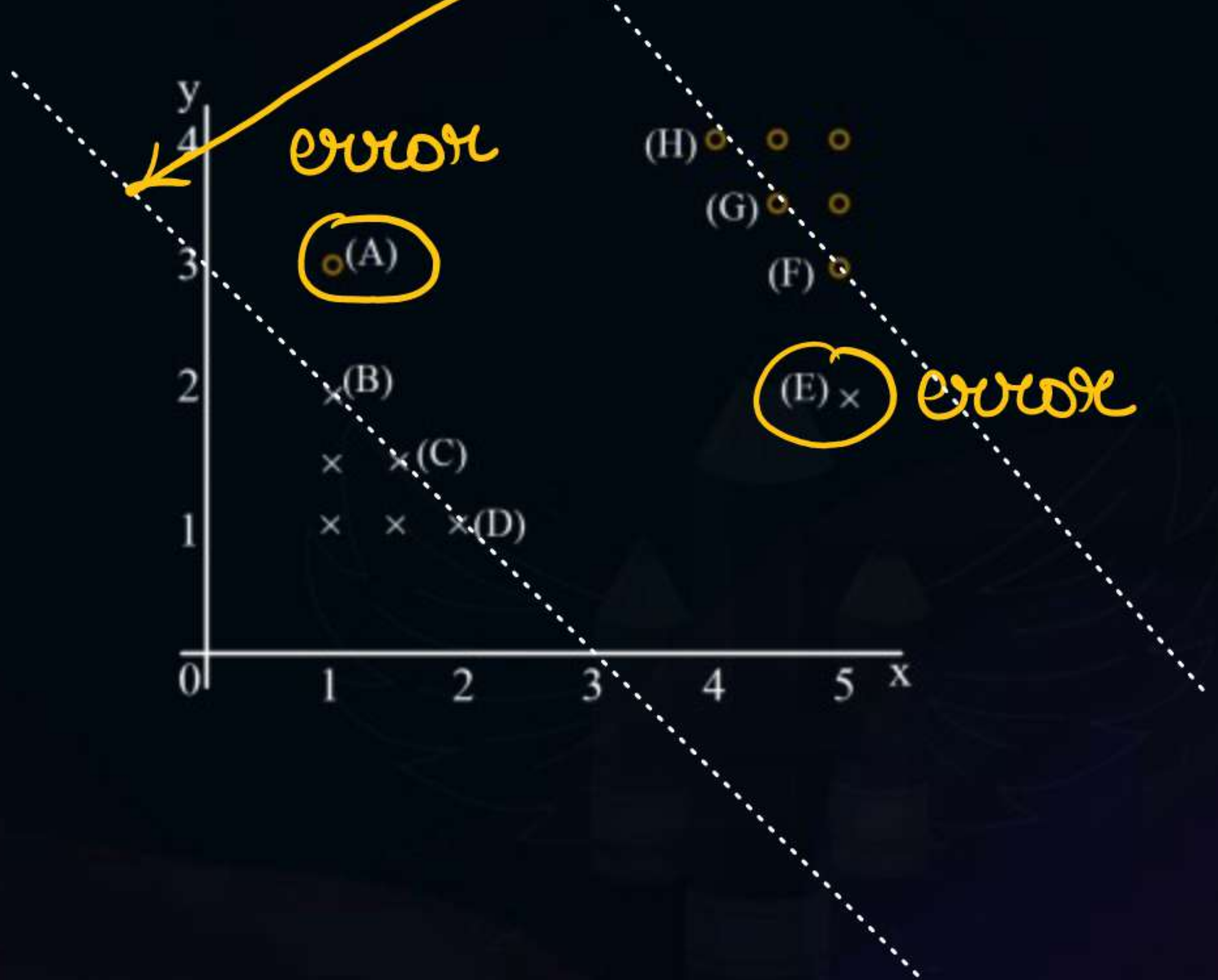
B ABGH

C ADEH

D ADEG

[MCQ]

#Q. Select all points that are support vectors. Soft margin



A ABEF

B ABGH

C ADEH

D ADEG

[MCQ]

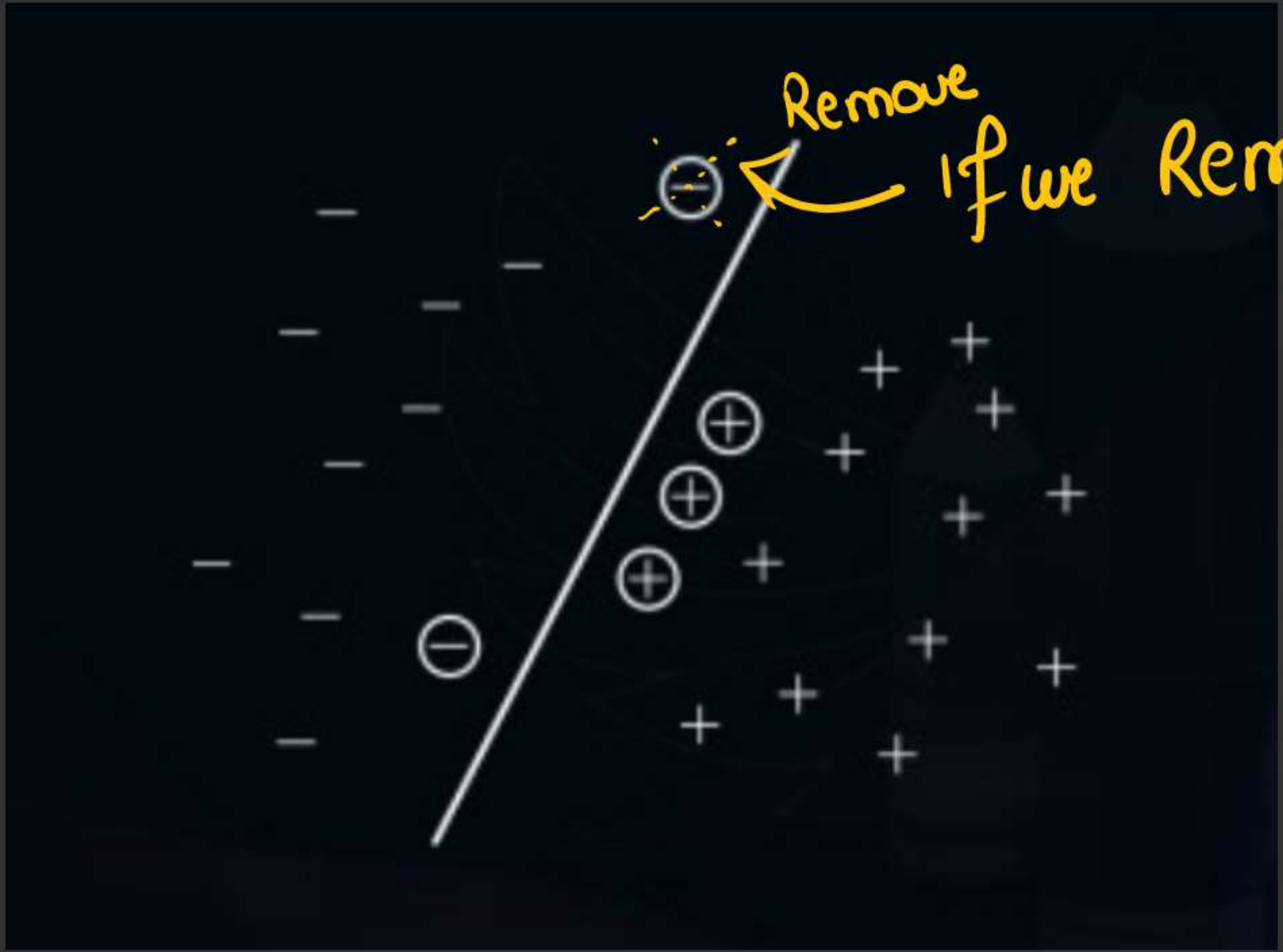
#Q. Consider a binary classification problem in a 2-dimensional instance space X = R^2. You are given a linearly separable training set. You run the hard-margin SVM algorithm and obtain the separating hyperplane below (support vectors are circled): What is the smallest number of data points that would have to be removed from the training set in order for the SVM solution to change?

A 1

B 2

C 3

D 4



If we Remove Single point then also Svm solution Changes

[MCQ]

#Q. Which of the following statements is true regarding the slack variable $\xi^{(i)}$ in SVM classification?

$0 < \xi < 1$: Correct Classify, but point is within margin

$\xi > 1$: misclassify,

A. $\xi^{(i)} > 0$ indicates that the point is classified correctly but lies inside the margin.

B. $\xi^{(i)} = 0$ indicates that the point lies on the decision boundary.

C. $\xi^{(i)} > 1$ indicates that the point is misclassified.

D. $\xi^{(i)} < 1$ indicates that the point is classified correctly but outside the margin.

THANK - YOU